DAY 18 – Data Exporting and Result Logging

Day 18 shifted focus from modeling toward **documentation and result management**, a crucial part of any data science project. Today's work involved saving predictions, exporting plots, and structuring final results for reporting. This ensures that every step is reproducible, organized, and stored for future reference.

🔹 1. Importance of Data Exporting

We learned why exporting output matters:

• Ensures work is not lost
• Stores predictions for future comparison
• Helps build professional reports
• Makes results shareable
• Improves reproducibility

Good documentation enhances project transparency.

🔸 2. Exporting Forecasts

The following exports were created:

✓ Future predictions saved as CSV files
✓ Cleaned datasets exported for later use
✓ Model summaries stored for documentation
✓ Confidence intervals exported for analysis

These files form the backbone of the final project.

🔸 3. Saving Visualizations

We saved several essential charts:

• Historical price trends
• Prediction vs actual comparison
• Confidence interval forecasts
• Error and residual plots

These graphs improve clarity in reports and presentations.

🔸 4. Logging Model Performance

We recorded:

• MAE, MSE, RMSE results
• Model configuration details
• Time taken to train the model
• Key observations from visual analysis

Logging allows tracking changes over time.

🔸 5. Organizing Project Files

Project files were neatly arranged:

• Data folder – raw + cleaned datasets
• Models folder – saved Prophet models
• Plots folder – exported figures
• Reports folder – summaries, notes, metrics
• Logs folder – recorded outputs

This structure reflects professional data-science workflow.

Key Takeaways from Day 18

• Exporting ensures long-term access to results
• Charts improve documentation quality
• Logging helps track model improvements
• Organized files reduce confusion
• Good documentation = Good project